

QUMARLEB, China

WMO: 560210

Lat: 34.1287N

Lon: 95.8060E

Elev: 4208

StdP: 59.99

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-22.5	-35.6	0.2	-15.9	-33.7	0.3	-13.7	12.7	1.1	10.8	-2.6	0.2	90	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	18.8	8.3	17.4	8.0	16.0	7.6	10.1	16.5	9.3	15.3	8.7	14.2	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
7.6	11.0	12.7	6.9	10.5	11.9	6.3	10.1	11.4	42.9	16.6	40.6	15.6	38.5	14.1	12.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.2	7.0	-28.3	21.7	1.7	1.6	-29.5	22.8	-30.5	23.8	-31.5	24.7	-32.7	25.9		
(5)			-28.8	11.5	1.6	0.5	-29.9	11.8	-30.9	12.2	-31.8	12.5	-32.9	12.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-12.1	-9.2	-5.4	-0.3	3.8	7.7	10.3	9.9	6.5	-0.1	-6.8	-11.0
	DBStd	8.47	3.66	3.65	3.33	3.03	2.59	2.64	2.30	2.49	2.77	3.62	3.20	3.35
	HDD10.0	3913	686	539	476	310	191	77	25	33	108	313	504	650
	HDD18.3	6879	944	772	734	560	449	319	250	262	354	571	754	908
	CDD10.0	76	0	0	0	0	1	8	33	30	4	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	2.3	2.2	2.8	2.9	2.8	2.7	2.4	2.2	2.1	2.0	2.1	1.9	1.9
(15) Precipitation	PrecAvg	407	3	3	7	11	38	81	95	77	70	17	3	2
	PrecMax	522	7	8	15	26	94	142	173	143	122	64	10	7
	PrecMin	288	0	0	1	2	7	13	37	29	35	2	0	0
	PrecStd	60	2	2	3	6	17	29	29	27	23	14	3	2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	5.0	9.2	13.1	17.7	18.6	21.1	21.4	17.6	14.2	5.9	3.7
		MCWB	-5.0	-4.2	-1.9	0.6	4.1	7.5	9.6	8.2	7.6	3.5	-3.1	-4.8
	2%	DB	0.6	2.4	6.1	10.6	14.2	16.5	18.8	19.0	15.6	10.4	4.0	1.0
		MCWB	-6.6	-5.4	-3.2	-0.1	3.1	7.3	8.8	8.5	7.4	2.2	-4.0	-6.3
	5%	DB	-1.9	0.6	4.2	8.8	12.1	14.8	17.4	17.4	13.7	8.1	2.2	-1.1
		MCWB	-8.0	-6.4	-4.0	-0.9	2.3	6.7	8.4	8.3	6.6	0.9	-4.8	-7.4
	10%	DB	-4.1	-1.1	2.4	6.9	10.2	13.2	15.9	15.7	12.0	6.2	0.4	-3.0
		MCWB	-9.1	-7.2	-4.8	-1.7	1.7	6.1	7.8	7.8	5.8	-0.1	-5.6	-8.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.8	-3.8	-1.4	2.0	6.1	10.0	11.1	10.8	9.2	5.2	-2.3	-4.4
		MCDB	2.6	3.6	6.8	9.3	12.7	15.7	17.8	17.1	15.2	12.1	3.8	2.8
	2%	WB	-6.4	-5.1	-2.6	0.7	4.5	8.7	10.2	9.9	8.2	3.5	-3.5	-6.1
		MCDB	0.0	1.5	5.0	8.4	10.8	14.3	16.8	16.2	13.9	8.5	2.5	0.3
	5%	WB	-7.7	-6.1	-3.6	-0.2	3.6	7.8	9.6	9.2	7.4	2.0	-4.4	-7.2
		MCDB	-2.3	0.1	3.4	6.9	9.5	12.9	15.7	15.2	12.5	6.6	1.4	-1.4
	10%	WB	-9.0	-7.1	-4.5	-1.1	2.7	7.0	8.8	8.5	6.5	0.7	-5.4	-8.3
		MCDB	-4.4	-1.5	1.8	5.5	8.3	11.8	14.3	14.1	10.8	4.6	-0.1	-3.3
(35) Mean Daily Temperature Range	MDBR		15.1	14.5	13.9	12.8	11.1	9.8	10.4	10.8	10.3	11.3	14.3	15.8
	5% DB	MCDBR	17.9	17.1	16.8	15.8	15.1	12.3	13.0	13.7	12.8	13.6	16.4	18.2
		MCWBR	12.5	11.3	9.8	7.7	6.6	5.3	5.2	5.3	5.8	6.5	9.7	12.5
	5% WB	MCDBR	16.8	15.8	15.0	13.4	11.5	9.7	10.4	10.5	10.7	11.5	14.8	17.7
		MCWBR	11.9	10.5	8.9	6.6	5.5	4.6	4.5	4.4	5.1	5.8	9.0	12.2
(40) Clear-Sky Solar Irradiance	taub		0.186	0.211	0.259	0.299	0.311	0.297	0.286	0.280	0.258	0.234	0.196	0.179
	taud		2.612	2.540	2.399	2.293	2.342	2.496	2.591	2.599	2.627	2.617	2.635	2.674
	Ebn at Noon		1077	1067	1024	986	971	980	989	995	1012	1025	1055	1075
	Edh at Noon		72	88	111	130	126	108	98	95	88	80	70	64
(44) All-Sky Solar Radiation	RadAvg		3.30	4.14	5.31	6.23	6.31	5.89	6.04	5.49	4.82	4.24	3.77	3.19
	RadStd		0.30	0.21	0.28	0.38	0.43	0.39	0.39	0.41	0.30	0.22	0.14	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.64	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.46	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page